Andhra University College of Engineering

Chemical Engineering | Course Code: CH301

Chemical Reaction Engineering - Comprehensive Syllabus Lecture Notes

Syllabus Focus: CH301 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Chemical Reaction Engineering. Includes detai

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Kinetics of Homogeneous Reactions

" Syllabus Scope: Rate of reaction, Temperature dependency, Arrhenius law, Collision theory, Transition s

Unit 2: Ideal Reactor Design Equations

" Syllabus Scope: Design equations for Batch, Continuous Stirred Tank CSTR, and Plug Flow Reactors PFR un

CH301: Chemical Reaction Engineering - Advanced Topics & Assessment

Academic Track: Chemical Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Multiple Reactor Networks & Yield

" CSTRs in series and parallel, Comparison of reactor volumes for first and second order reactions, Product se

Unit 4: Non-Isothermal Reactor Design

" Energy balance for batch, CSTR, and PFR, Adiabatic operations, Equilibrium conversion, Optimum temperature p

Unit 5: Non-Ideal Flow & Residence Time Distribution

" Residence time distribution RTD, Et and Ft curves, Pulse and step tracer stimulus experiments, Tank-in-serie

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Derive the performance equation for an isothermal Plug Flow Reactor PFR for a second-order liquid phas
2. Compare the total volume required for two equal CSTRs in series vs a single CSTR for 90% conversion of
3. Formulate the Et residence time distribution function from experimental step tracer response data.

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* H. Scott Fogler - Elements of Chemical Reaction Engineering Prentice Hall 5th Ed
* Octave Levenspiel - Chemical Reaction Engineering Wiley 3rd Ed

Authorized NoteVault Educational Document " Verified for Registered University Students